Bottom heaviness of the four q-Stirling families

A solution to MathDB problem 354788

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Task 1. Solve the original conjecture. For a nonzero polynomial $F(q) = \sum_{i=0}^{D} a_i q^i$ of degree D, bottom heaviness means $a_i \geq a_{D-i}$ for $0 \leq i < D/2$. We prove this for all four polynomials in Conjecture 7.6 of Sagan–Swanson [1]. Constants and zero polynomials satisfy the condition vacuously.

Put $[m]_q = 1 + q + \cdots + q^{m-1}$ for integers $m \geq 1$, and $[0]_q = 0$. The source's four families have initial values $S[0,k] = c[0,k] = S_B[0,k] = c_B[0,k] = \delta_{0k}$ and recurrences

$$S[n,k] = S[n-1,k-1] + [k]_q S[n-1,k], \tag{1}$$

$$c[n,k] = c[n-1,k-1] + [n-1]_q c[n-1,k], \tag{2}$$

$$S_B[n,k] = S_B[n-1,k-1] + [2k+1]_q S_B[n-1,k], \tag{3}$$

$$c_B[n,k] = c_B[n-1,k-1] + [2n-1]_q c_B[n-1,k]. \tag{4}$$

Here $n \geq 1$, $0 \leq k \leq n$, and entries outside the triangular range are zero. These are equations (2), (7), (12), and (14) of [1]. In particular, $S[n,k]$ is the unshifted version in (2).

Lemma 1. *Every finite product of q-integers $[m]_q$, $m \geq 1$, has nonnegative, symmetric, unimodal coefficients.*

Proof. We give an elementary induction that also handles plateaux. Suppose $f(q) = \sum_{j=0}^{d} f_j q^j$ has the asserted properties, and set $f_j = 0$ for all integers outside $[0,d]$. The value f_j is a nonincreasing function of $|j - d/2|$ on these integer indices. For $g = f[m]_q$, symmetry follows by reversing both factors. Moreover,

$$g_i - g_{i-1} = f_i - f_{i-m}.$$

If $1 \leq i \leq (d+m-1)/2$, then

$$(i - m - d/2)^2 - (i - d/2)^2 = m(d + m - 2i) \geq 0.$$

Thus $f_i \geq f_{i-m}$, so g is nondecreasing up to its center $(d+m-1)/2$. Symmetry gives unimodality. Induction starts with the empty product 1. □

Lemma 2. *A finite nonnegative linear combination of nonzero polynomials having nonnegative, symmetric, unimodal coefficients is bottom heavy.*

Proof. Discard zero weights. If all weights vanish the assertion is vacuous; otherwise let D be the largest degree of a summand. Nonnegativity ensures that D is the degree of the sum. Fix a summand f of degree $d \leq D$, extend its coefficient sequence by zeros, and let $0 \leq i < D/2$. Since

$$(D - i - d/2)^2 - (i - d/2)^2 = (D - 2i)(D - d) \geq 0,$$

the same distance-from-center observation gives $f_i \geq f_{D-i}$. Multiplying by the nonnegative weight and adding over the summands proves the claim. □

Theorem 3. *For every admissible n, k, the coefficient sequences of $S[n,k]$, $c[n,k]$, $S_B[n,k]$, and $c_B[n,k]$ are bottom heavy. More generally, every finite nonnegative linear combination of products of q-integers is bottom heavy.*

Proof. The general assertion is the combination of Lemmas 1 and 2. Let $\mathcal{C}$ consist of these linear combinations, including 0 and the empty product 1. This class is closed under addition and multiplication by any $[m]_q$, $m \geq 0$. Every coefficient multiplying an entry in (1)–(4) is such a q-integer. Induction on n therefore places every entry in $\mathcal{C}$. This also handles $k = 0$, $k = n$, and $n = 0$ without a positivity or degree assumption on a zero entry. □

The comparison in Lemma 2 uses each summand's own center: the summands need not have equal degrees. For example $S[4,3] = [1]_q + [2]_q + [3]_q = 3 + 2q + q^2$ has the required asymmetry. The theorem settles the entire original conjecture; it does not assert unimodality of a sum with different centers.

Verification. The proof is independent of computation. The accompanying exact integer checker reconstructs all four recurrences for $n \leq 50$, compares them with explicit complete-homogeneous and elementary symmetric-function expansions for $n \leq 10$, and tests the lemmas on products and mixtures with unequal degrees. It also reproduces the source's nonunimodal type B examples.

Task 2. Construct coefficient injections. Translate the summand comparison into an explicit injection from objects counted by the coefficient of q^{D-i} to those counted by q^i, using products of chains and the source's partition and permutation models.

Task 3. Classify equality and quantify the asymmetry. Determine exactly when $a_i = a_{D-i}$ in each family, and obtain effective lower bounds for $a_i - a_{D-i}$ outside those cases.

Task 4. Extend the theorem to further Stirling analogues. Identify which q-Stirling polynomials for complex reflection groups have nonnegative expansions in products of q-integers, then apply the theorem and characterize any exceptions.

Task 5. Resolve the stronger coefficient conjectures. Prove or refute coefficient log-concavity in type A and parity log-concavity in type B, as stated in Conjectures 7.4 and 7.5 of [1]; isolate additional structure beyond the sum lemma.

References

[1] B. E. Sagan and J. P. Swanson, *q-Stirling numbers in type B*, European Journal of Combinatorics **118** (2024), 103899. Published text; arXiv:2205.14078. Definitions: pp. 2–4; Conjecture 7.6: p. 31.